

1. Convert Celsius to Fahrenheit

```
#!/bin/bash

# Ask the user to enter temperature in Celsius
echo "Enter temperature in Celsius:"
read celsius

# Validate if input is a number (can include decimals)
if ! [[ "$celsius" =~ ^-?[0-9]+(\.[0-9]+)?$ ]]; then
    echo "Invalid input. Please enter a numeric value."
    exit 1
fi

# Convert to Fahrenheit using formula:  $F = (C \times 9/5) + 32$ 
fahrenheit=$(echo "scale=2; ($celsius * 9 / 5) + 32" | bc)

# Display the result
echo "$celsius°C is equal to $fahrenheit°F"
```

2. Shell script that takes a person's age as input and prints their age group based on your criteria:

```
#!/bin/bash

# Ask the user for their age
echo "Enter your age:"
read age

# Validate input: Check if age is a non-negative integer
if ! [[ "$age" =~ ^[0-9]+$ ]]; then
    echo "Invalid input. Please enter a valid non-negative number."
    exit 1
fi
```

```
# Determine age group
if [ "$age" -ge 0 ] && [ "$age" -le 12 ]; then
    echo "Age group: Child"
elif [ "$age" -ge 13 ] && [ "$age" -le 19 ]; then
    echo "Age group: Teenager"
elif [ "$age" -ge 20 ] && [ "$age" -le 64 ]; then
    echo "Age group: Adult"
else
    echo "Age group: Senior"
fi
```

3. A movie theatre offers different ticket prices based on age:

- Child (0-12 years): \$8
- Adult (13-64 years): \$12
- Senior (65+ years): \$6

Write a shell script that calculates the movie ticket price based on the user's age.

```
#!/bin/bash

# Ask the user for their age
echo "Enter your age:"
read age

# Validate input: Check if age is a non-negative integer
if ! [[ "$age" =~ ^[0-9]+$ ]] || [ "$age" -lt 0 ]; then
    echo "Invalid input. Please enter a valid non-negative number."
    exit 1
fi

# Determine the ticket price based on age group
if [ "$age" -ge 0 ] && [ "$age" -le 12 ]; then
    price=8 # Child ticket
elif [ "$age" -ge 13 ] && [ "$age" -le 64 ]; then
    price=12 # Adult ticket
else
    price=6 # Senior ticket
fi

# Output the calculated ticket price
echo "The movie ticket price is: \${price}"
```